Computer Networks

Lecture- 08

Recap of Lecture 07

- Transmission Media and its Types
- Guided Media
  - Twisted Pair
    - UTP
    - STP
  - Coaxial Cable
  - Optical Fiber
    - Multimode
    - Single mode

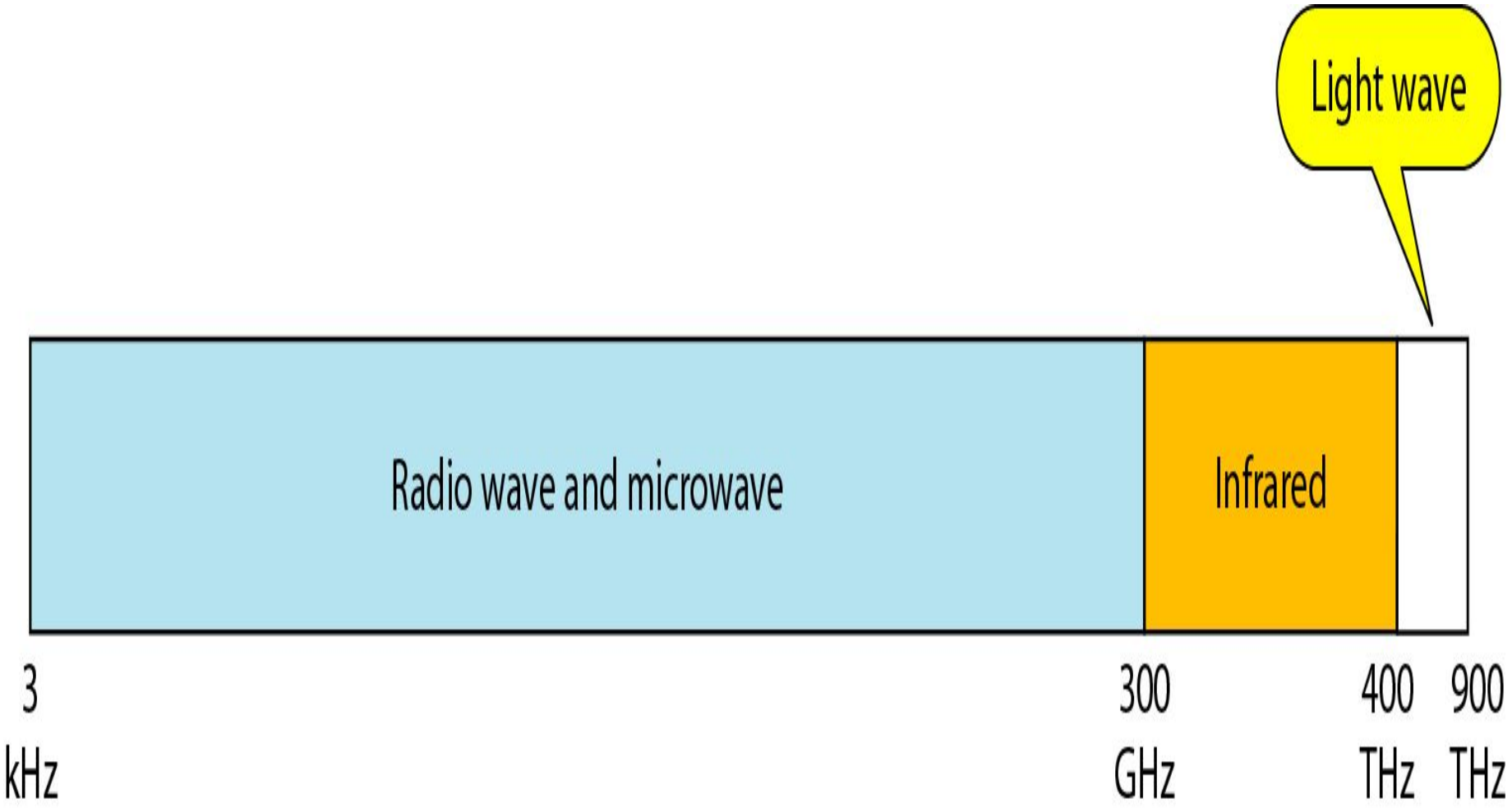
Overview of Lecture

- Unguided Media
- Propagation methods
- Radio Frequency Allocation
  - Radio Waves
  - Microwave
  - Infrared
- Media Comparison

Unguided Media

- Unguided media transport electromagnetic waves without using a physical conductor.
- This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

Electromagnetic spectrum for wireless communication



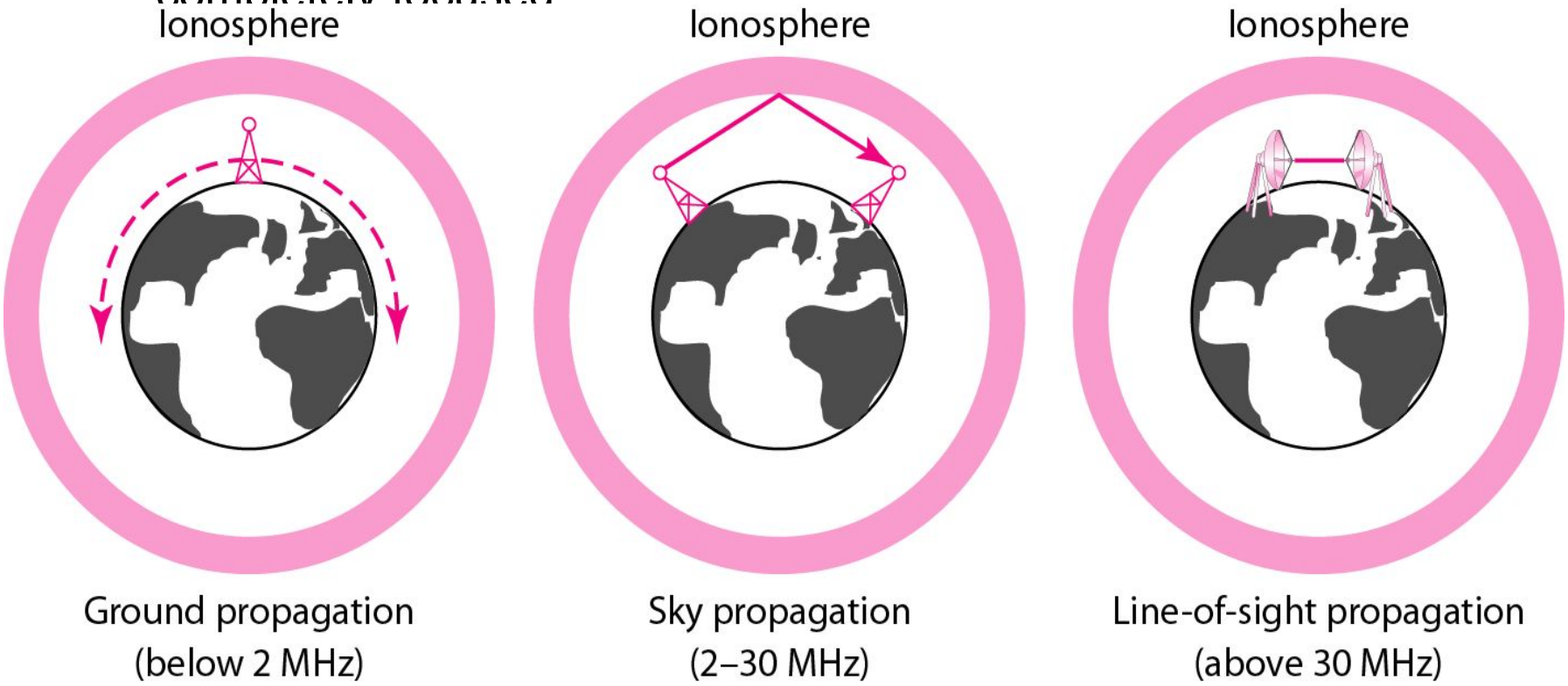
Propagation methods

- Unguided signals can travel from the source to destination in several ways:
  1. Ground propagation,
  2. Sky propagation, and
  3. Line-of-sight propagation

- **Ground propagation,**
  - Radio waves travel through the lowest portion of the atmosphere, hugging the earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance.
- **sky propagation**,
  - higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power.

Propagation methods

- **line-of-sight propagation**,
  - very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth. Line-of-sight propagation is tricky because radio transmissions cannot be completely focused



Radio Frequency Allocation

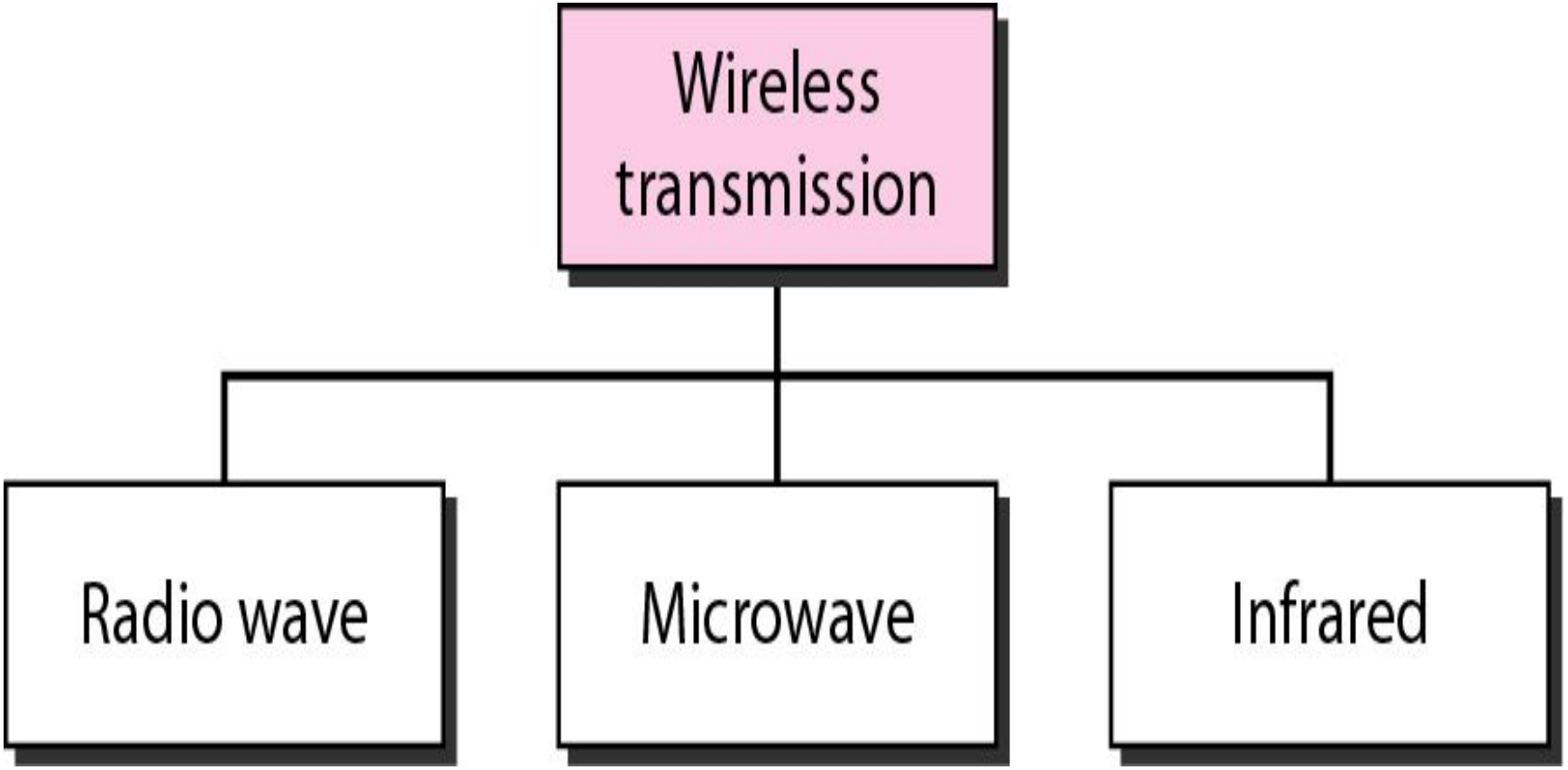
- The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges, called bands, each regulated by government authorities.

- These bands are rated from very low frequency (VLF) to extremely high-frequency (EHF).

Bands

Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

Wireless Transmission



Radio wave

- Although there is no clear-cut demarcation between radio waves and microwaves, electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 and 300 GHz are called microwaves.

- Omni-directional (all direction)
- Sending and receiving antennas do not have to be aligned.
- The omnidirectional property has a **disadvantage, too**.
  - The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.

Radio wave

- Radio waves, particularly those waves that propagate in the sky mode, can travel long distances (therefore a good choice to AM radio broadcasting).
- Low and medium frequency radio waves can penetrate walls.
- This characteristic can be advantage and disadvantage

  - **Advantage:**
    - AM radio signals can be received inside a building

  - **Disadvantage:**
    - Cannot be limited in a single building.

Application

- The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers.
  1. AM and FM radio,
  2. television, maritime radio,
  3. cordless phones, and
  4. Paging and so on..

Microwave

Microwave

- Electromagnetic waves having frequency between 1GHz and 300GHz.
- Microwaves do not follow the curvature of earth.
- Unidirectional waves, means sending and receiving antennas need to be aligned.

- Line-of-sight transmission.
- Height allows the signal to travel farther.
- Very high frequency microwaves cannot penetrate walls.
- For long distance repeaters are used.

Unidirectional antenna

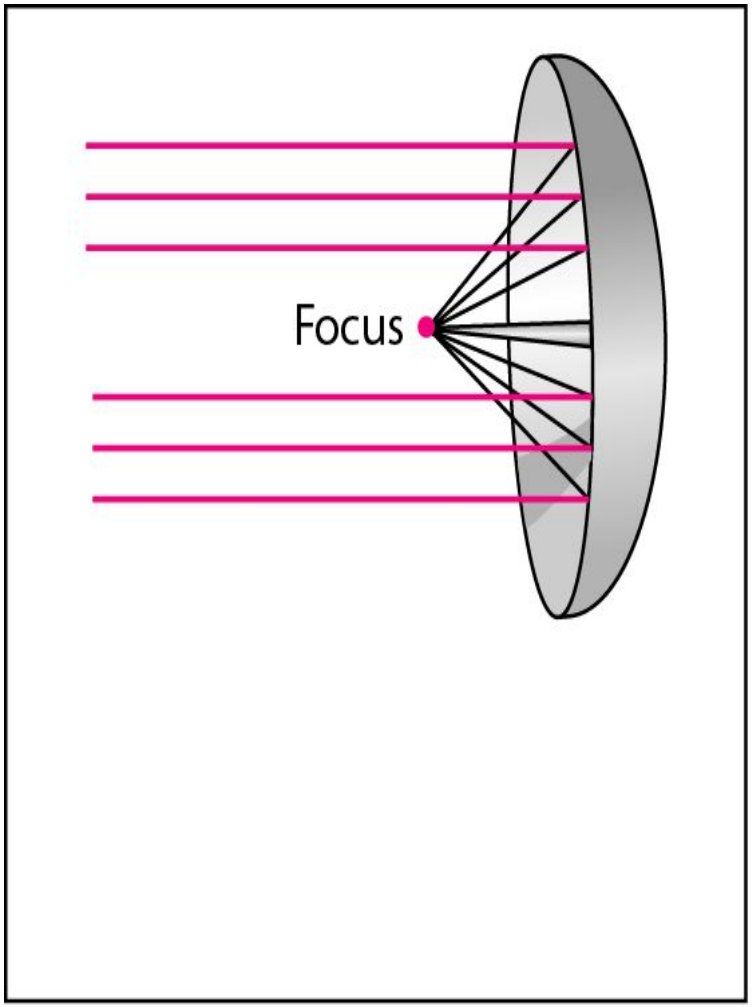
- Microwaves need unidirectional antennas that send out signals in one direction.
- Two types of antennas are used for microwave communications:
  1. Parabolic dish Antenna
  2. Horn Antenna

- **Parabolic dish antenna**
  - It is based on the geometry of a parabola:
  - Every line parallel to the line of symmetry (line of sight) reflects off the curve at angles such that all the lines intersect in a common point called the focus.
  - The parabolic dish works as a funnel, catching a wide range of waves and directing them to a common point.
  - In this way, more of the signal is recovered than would be possible with a single-point receiver.

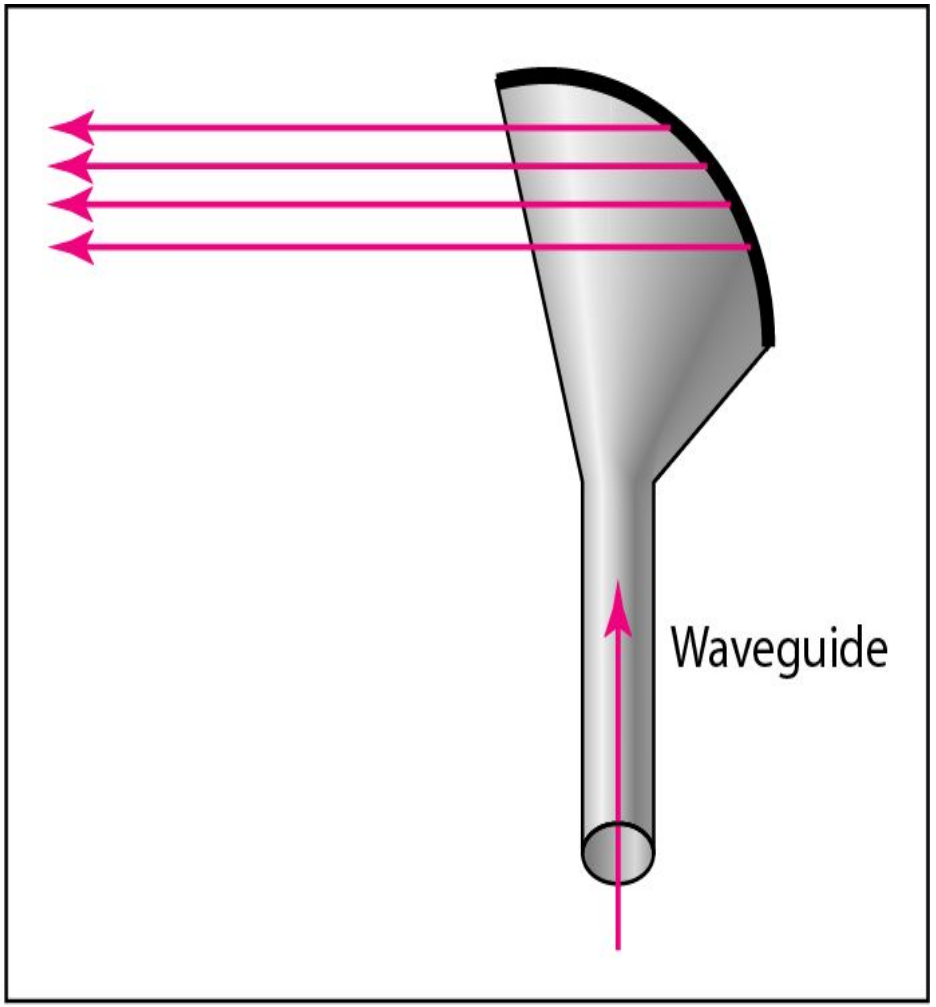
Unidirectional antenna

- **Horn Antenna**
  - Outgoing transmissions are broadcast through a **horn** aimed at the dish.
  - The microwaves hit the dish and are deflected outward in a reversal of the receipt path.
  - A horn antenna looks like a gigantic scoop.
  - Outgoing transmissions are broadcast up a stem (resembling a handle) and deflected outward in a series of narrow parallel beams by the curved head.
  - Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.

Unidirectional antenna

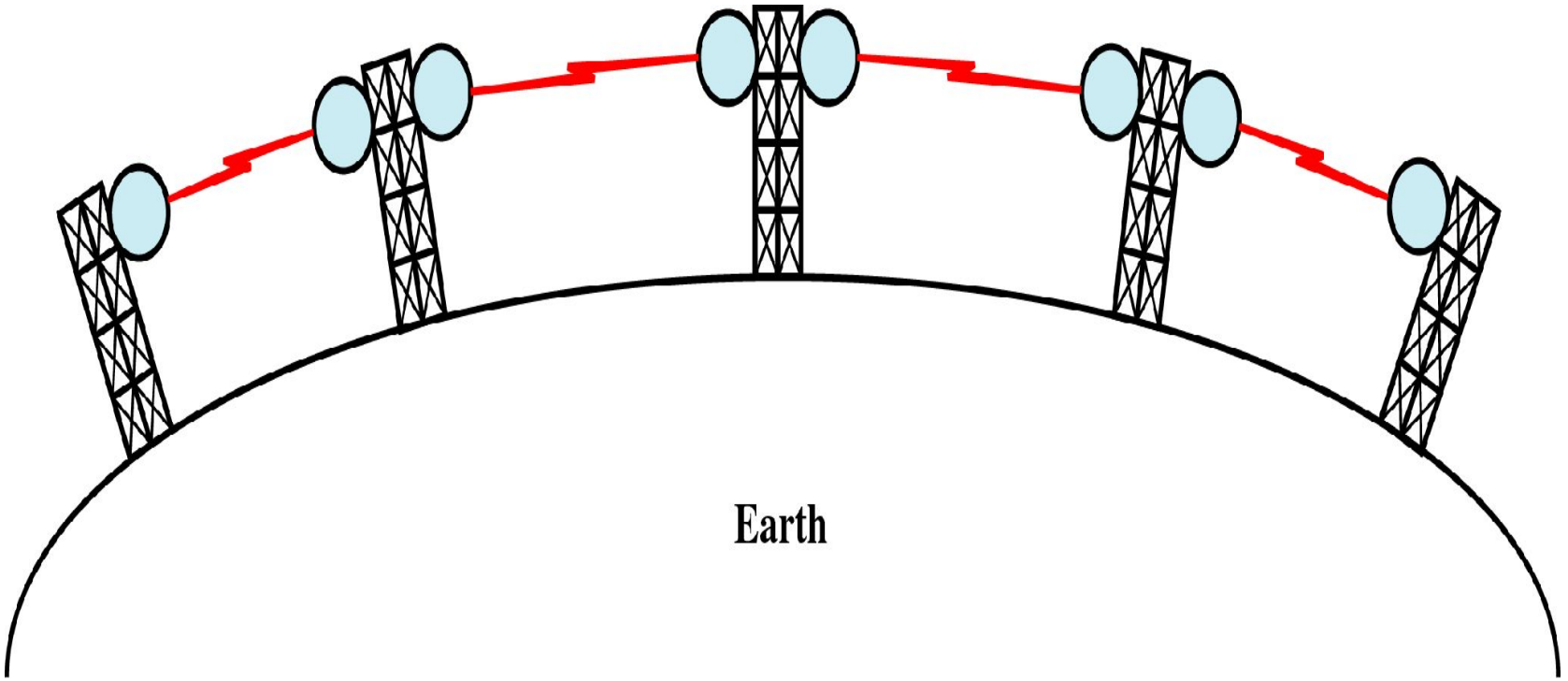


a. Dish antenna



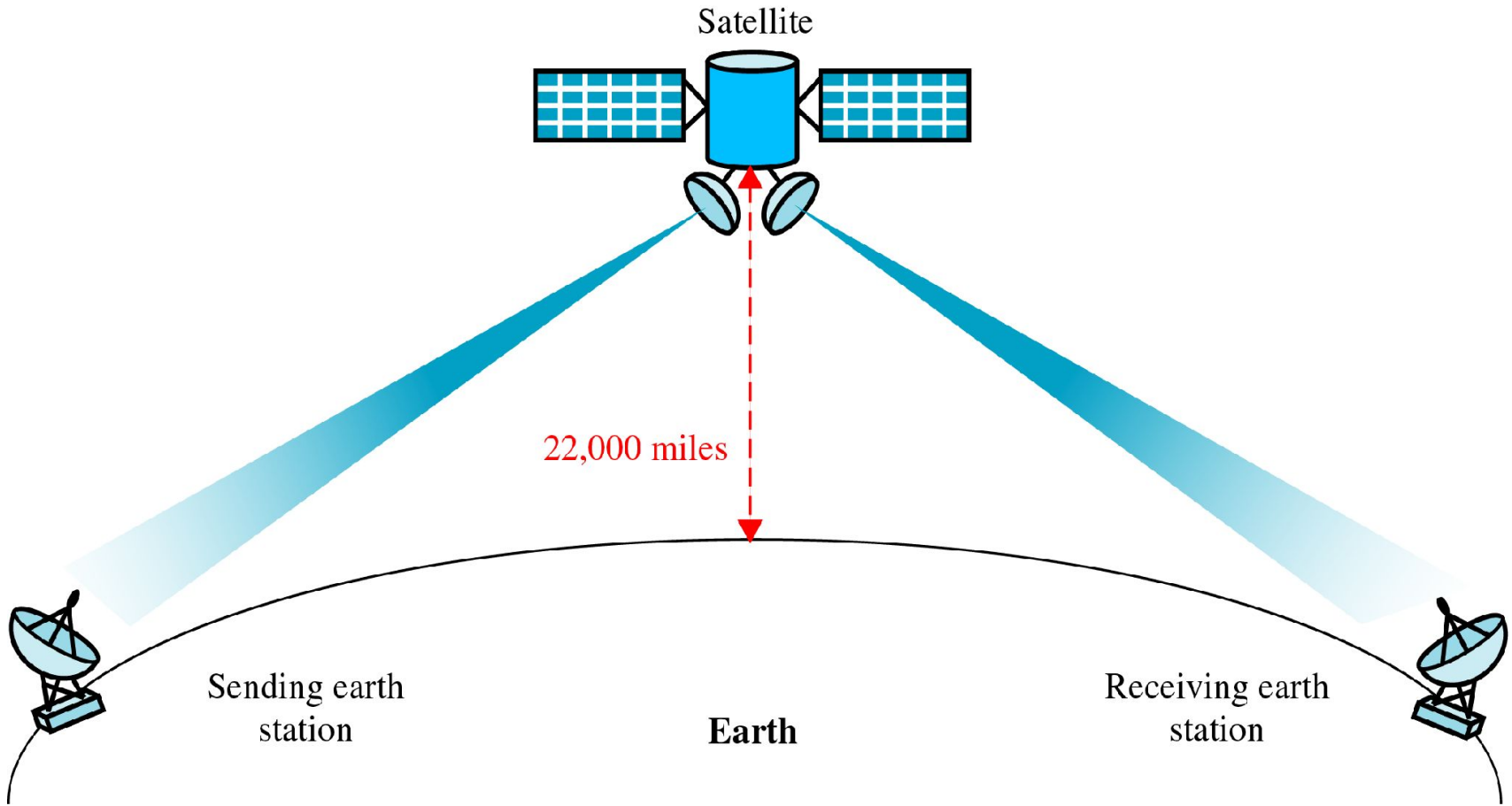
b. Horn antenna

Terrestrial Microwave



Satellite Communication

- Line-of-sight microwave communication using satellite.
- Satellite acts as a very tall antenna and a repeater.



Applications

- Microwaves, due to their unidirectional properties, are very useful when unicast (one-to-one) communication is needed between the sender and the receiver.

- They are used in;
  1. Cellular phones,
  2. Satellite networks, and
  3. Wireless LANs

Infrared

Infrared

- Frequency is 300 GHz to 400 GHz.
- Used for short range communication.
- Due to high frequency, cannot penetrate walls (advantage: one system cannot effect other).
- E.g. TV remote control
- Useless in long range transmission.
- Cannot be used in sun light, as it also contain infrared rays which makes interference.
- data rate of 75Kbps for 8m distance, but nowadays 4Mbps.

Applications

- The infrared band, almost 400 THz, has an excellent potential for digital data transmission with very high data rate.
- The **Infrared Data Association (IrDA),** an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers. For example, some manufacturers provide a special port called the IrDA port that allows a wireless keyboard to communicate with a PC.
- The standard originally defined a data rate of 75 kbps for a distance up to 8 m.
- The recent standard defines a data rate of 4 Mbps.
- Infrared signals defined by IrDA transmit through line of sight; the IrDA port on the keyboard needs to point to the PC

Media Comparison

- Five factors should be kept in mind when evaluating a medium:

1. Cost
2. Speed
3. Attenuation
4. EM Interference
5. Security

Summary of Lecture 20

- Unguided Media
- Propagation methods
- Radio Frequency Allocation
- Radio Waves
- Microwave
- Infrared
- Media Comparison

Suggested Reading

- Section
  - 7.2
    - “Data Communications and Networking” 4th Edition by Behrouz A. Forouzan